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Analysis and robust control of system with uncertainty and disturbance using equivalent-input-disturbance approach

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Abstract

This paper addresses the control problem of an uncertain system suffering from an exogenous disturbance. A new degree of control freedom is developed to handle the problem based on the equivalent-input-disturbance (EID) approach. The effect of the disturbance and uncertainties is equivalent to that of a fictitious disturbance on the control input channel, which is called an EID. A state observer and an improved EID (IEID) estimator are devised to produce an estimate that is used to compensate for the disturbance and uncertainties in a control law. A second-order low-pass filter is employed in the estimator to provide a way to solve a tradeoff between disturbance rejection and noise suppression. The slope of the Bode magnitude curve at high frequencies is two times larger for the IEID estimator than for a conventional one. This makes the IEID estimator less sensitive to measurement noise and more practical. Sufficient analyses reveal the mechanism of disturbance rejection, uncertainty attenuation, and noise suppression of an IEID-based control system. A theorem is derived to guarantee system stability and a procedure is presented for system design. Simulations and experiments of the position control of a magnetic levitation system are carried out to show the validity of the presented method.

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1. Introduction

Many industrial control systems suffer from exogenous disturbances and uncertainties [1,2], for example, force disturbances from environments and parameter variations caused by age deteriorations and temperature fluctuation [3]. Disturbances and uncertainties degrade system performance and even deteriorate system stability. One of the essential properties of a control system is the robustness against disturbances and uncertainties [4]. The proportional-integral-derivative (PID) control [5], H_∞ control [6], and many other feedback control methods have been proposed to enhance the robustness of a control system. However, these methods handle disturbances and uncertainties in a passive way by expanding the control bandwidth of a system. They have only one degree of control freedom. The design of a controller based on these methods should consider the specifications of both reference tracking and disturbance rejection.

Several elegant approaches that have two degrees of control freedom have been presented to do that in an active way. They treat unknown disturbances and plant uncertainties as a lumped disturbance. One degree is to construct an inner loop to estimate and compensate for the lumped disturbance actively. The other is to force a system to achieve desired characteristics [7]. The active disturbance rejection control (ADRC) [8,9] is a popular method among these approaches. It uses only the minimum information on a plant (i.e., the relative degree of a plant from the control input to the output) to calculate an estimate of a disturbance. Xue and Huang provided sufficient analyses of the ADRC for uncertain systems in both the time and the frequency domains [10]. Sun et al. proposed a composite control design for uncertain systems with noise using the ADRC and a Kalman filter [11]. The disturbance observer-based control (DOBC) [12,13] is another commonly used method. It employs an inverse model of a plant to estimate the lumped disturbance. Chen et al. presented a principle of designing DOBC for a class of nonlinear uncertain systems [14]. A systematic view of these two methods was provided in several excellent books (for example, [15,16]) and a recent review of them was presented in [17–19]. The equivalent-input-disturbance (EID) approach focuses on the effect of a disturbance on the output of a system [20,21]. It integrates the information about a control input, a system output, and a state observer to produce an estimate of an equivalent disturbance on the control input channel. The equivalent disturbance, which is also called an EID, has the same effect on the system output as the actual disturbance does. A deep observation and comparison of the ADRC, DOBC, and the EID approach is given in [22].

The EID approach is devoted to calculating an estimate of an EID rather than a disturbance itself. This removes constraints on the type and the imposed channel of a disturbance. It has been extensively studied and applied to feed drive control systems [23], motion control systems [24,25], time-delay systems [26], and many other industrial systems [27,28]. Although it has been extended to uncertain systems [29,30], only the analysis of system stability and disturbance rejection was investigated. So far, there is insufficient analysis of the uncertainty-attenuation mechanism of the EID approach in the literature. Moreover, few studies have addressed the tradeoff between disturbance rejection and noise suppression [31,32]. Wu et al. incorporated a proportional-integral observer into an EID estimator to filter out measurement

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noise [31]. Yin et al. used two low-pass filters in an EID estimator to suppress high-frequency components in system output [32].

This paper presents an improved EID (IEID) approach to reject disturbances and attenuate parameter uncertainties. First, an inner loop that consists of a state observer and an IEID estimator is constructed to obtain satisfactory performance of disturbance rejection and uncertainty attenuation. Next, a second-order low-pass filter is employed in the estimator to improve the noise-suppression performance of the inner loop. Then, the mechanism of disturbance rejection, uncertainty attenuation, and noise suppression is analyzed adequately in the frequency domain. Stability criteria are derived and a design procedure is provided. Finally, the validity and the superiority of the method are demonstrated by the simulation and experimental results of the position control of a magnetic levitation system (MLS). The main contribution of the paper is as follows.

- (1) The existence of an EID for an uncertain system is first proven explicitly.
- (2) An IEID estimator that employed a second-order low-pass filter is presented to solve the tradeoff between disturbance rejection and noise suppression in the inner loop.
- (3) Sufficient analyses are provided to show the mechanism of disturbance rejection, uncertainty attenuation, and noise suppression of the presented method. Consequently, a design procedure for an IEID-based control system is given.

The remainder of the paper is organized as follows. [Section 2](#) presents problem formulation and describes the configuration of an IEID estimator. [Section 3](#) provides sufficient analyses to explain the mechanism of disturbance rejection, uncertainty attenuation, and noise suppression. Furthermore, it analyzes system stability and gives a design procedure for an IEID-based control system. [Section 4](#) demonstrates simulation and experimental results. [Section 5](#) provides some concluding remarks.

2. Problem formulation and configuration of IEID estimator

Consider the following uncertain plant

$$\begin{cases} \dot{x}(t) = (A_n + \Delta A)x(t) + (B_n + \Delta B)u(t) + B_d d(t) \\ y(t) = Cx(t) \end{cases} \quad (1)$$

where $x(t)$, $y(t)$, and $u(t)$ are the state, the output, and the control input of the plant, respectively; A_n and B_n are nominal parameters; ΔA and ΔB are parameter uncertainties; $d(t)$ is an exogenous disturbance; and B_d is the parameter of a channel that the disturbance is imposed on. Assume

$$y(t) = y_n(t) + \Delta y(t) + y_d(t) \quad (2)$$

where $y_n(t)$ is the output of the plant (A_n, B_n, C) caused by the input $u(t)$, $\Delta y(t)$ is the output of the plant $(\Delta A, \Delta B, C)$ caused by the input $u(t)$, and $y_d(t)$ is the output of the plant $((A_n + \Delta A), B_d, C)$ caused by the disturbance $d(t)$. Many practical systems can be modeled as [Eq. \(1\)](#), for example, hard drive disk systems and multi-mass spring systems, which can be denoted by a rigid body plus resonant modes.

Assumption 1. The pair (A_n, B_n) is controllable and (A_n, C_n) is observable. The nominal plant (A_n, B_n, C_n) has no zeros on the right-half complex plain.

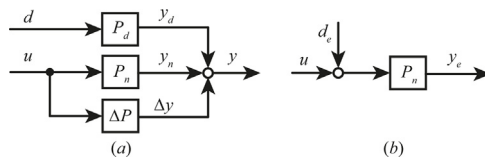


Fig. 1. Block diagrams of plants. (a) Plant (1). (b) Plant (3).

Assumption 1 is standard. If the nominal plant (A_n, B_n, C_n) is not controllable and/or observable, then we should consider its controllable and observable subsystem. The assumption that the nominal plant has no unstable zeros is used to allow the output to track a reference input without a steady-state error by using output regulation.

The disturbance and the parameter uncertainties are treated as a lumped disturbance. The effect of the lumped disturbance is equivalent to that of an EID. The following lemma is derived for the existence of the EID.

Lemma 1. *There exists a signal $d_e(t)$ on the control input channel of the plant*

$$\begin{cases} \dot{x}_e(t) = A_n x_e(t) + B_n [u(t) + d_e(t)] \\ y_e(t) = C x_e(t) \end{cases} \quad (3)$$

where $x_e(t)$ is the state and $y_e(t)$ is the output. For all $t \geq 0$, the output of Eq. (3) equals that of Eq. (1), that is

$$y_e(t) \equiv y(t), \quad t \geq 0. \quad (4)$$

The signal $d_e(t)$ is called an EID of the lumped disturbance.

Proof. Figure 1a and b show the block diagrams of the plant Eqs. (1) and (3), respectively. P_n and ΔP are the nominal and uncertainty units, respectively. P_d is the unit from the disturbance d to the output y_d . It is easy to obtain

$$y = P_n u + \Delta P u + P_d d, \quad y_e = P_n u + P_n d_e. \quad (5)$$

First, suppose Eq. (4) hold, then combining Eqs. (4) and (5) yields

$$\Delta P u + P_d d = P_n d_e. \quad (6)$$

If the relative degrees from $d(t)$ to $y_d(t)$ and from $u(t)$ to $\Delta y(t)$ are less than or equal to that from $u(t)$ to $y_n(t)$, then a causal solution of d_e is calculated by

$$d_e = \frac{\Delta P}{P_n} u + \frac{P_d}{P_n} d. \quad (7)$$

Conversely, if d_e in Fig. 1b is given by Eq. (7), then substituting it into Eq. (5) implies that Eq. (4) always holds. This completes the proof. $\square$

We use the plant Eq. (3) in the rest of the paper and abuse notations a bit by using $x(t)$ and $y(t)$ instead of $x_e(t)$ and $y_e(t)$, respectively. This should not cause confusion.

The control objective in this study is to minimize the effect of the disturbance and the parameter uncertainties on the output by manipulating the control input. According to Lemma 1, the main issue of achieving the objective is to calculate d_e . However, the calculation in Eq. (7) requires the information on the disturbance and the uncertainties. It is not available. In this paper, the EID approach is employed to estimate d_e . Figure 2 shows the conceptual

diagram of the EID approach. First, a state observer is used to reconstruct the system state. Then, an IEID estimator is designed to calculate an estimate of d_e based on the difference between the output of the plant and that of the observer. The state observer is

$$\begin{cases} \dot{\hat{x}}(t) = A_n \hat{x}(t) + B_n u_f(t) + L[y(t) - \hat{y}(t)] \\ \hat{y}(t) = C \hat{x}(t) \end{cases} \quad (8)$$

where $\hat{x}(t)$ is the state, $\hat{y}(t)$ is the output, $u_f(t)$ is the input, and L is the gain of the observer. Define

$$\Delta x(t) = x(t) - \hat{x}(t). \quad (9)$$

Following the idea in [33], one obtains an estimate

$$\hat{d}_e(t) = B_n^+ LC \Delta x(t) + u_f(t) - u(t) \quad (10)$$

where B_n^+ is the pseudo-inverse of B_n .

A first-order low-pass filter is used to select a disturbance-estimation frequency range in a conventional EID estimator [33]. It is chosen as

$$F_c(s) = \frac{1}{\tau s + 1} \quad (11)$$

where τ is the time constant. As stated in [34], the accuracy of disturbance estimation improves as τ decreases. Generally, τ is selected such that the cutoff frequency ($1/\tau$) of the filter is 5–10 times larger than the highest frequency of a disturbance. However, this could cause the estimator to be sensitive to measurement noise. In the IEID estimator, a new filter

$$F(s) = \frac{1}{\tau^2 s^2 + 1.41 \tau s + 1} \quad (12)$$

is employed to filter out high components of measurement noise. Figure 3 shows Bode diagrams for the filters with different values of τ in the EID and IEID estimators. The characteristics of the filters are almost the same within the bandwidth of them. The slope of the magnitude curve is -20 dB/dec for the filter $F_c(s)$ in the EID estimator and -40 dB/dec for the filter $F(s)$ in the IEID estimator at high frequencies.

Since an estimate $\tilde{d}_e(t)$ is already available, we devise an inner-loop compensation mechanism by incorporating it into the control law $u_f(t)$. This yields a new control input

$$u(t) = u_f(t) - \tilde{d}_e(t) \quad (13)$$

to the plant.

The disturbance and uncertainties are converted into an equivalent disturbance. The methodology used in this paper focuses on the equivalent disturbance rather than the disturbance and uncertainties themselves. Although the IEID estimator cannot classify the effect of the disturbance and that of uncertainties in the estimation process, it can suppress the total effect of them on the system output. In the following section, we will provide a deep analysis to reveal that the disturbance will be blocked to be transmitted to the output and the plant will behave like a nominal one without uncertainties.

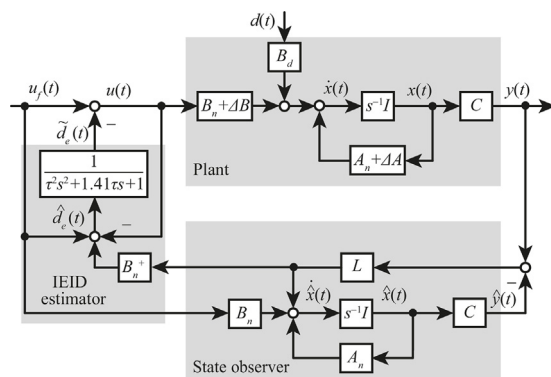


Fig. 2. Conceptual diagram of the EID approach.

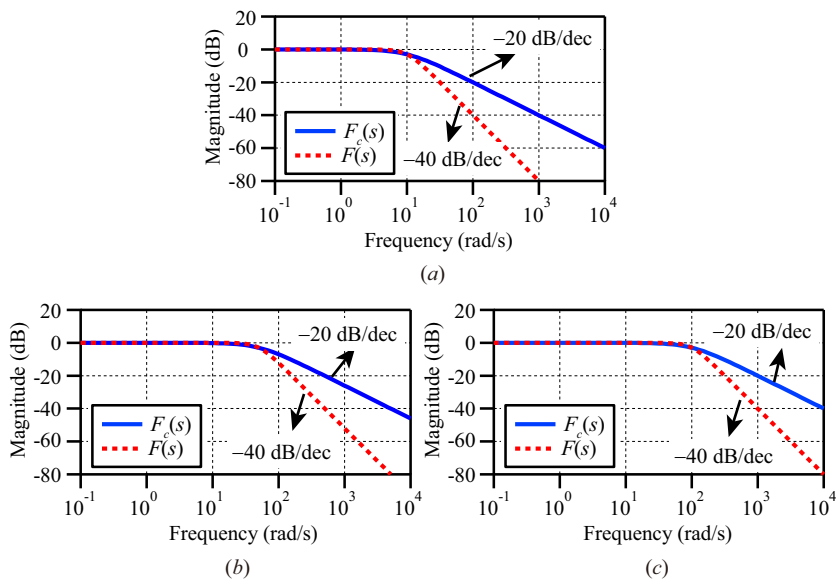


Fig. 3. Bode diagrams for filters in EID and IEID estimators: (a) $\tau = 0.1$, (b) $\tau = 0.02$, and (c) $\tau = 0.01$.

3. Performance analysis and stability of IEID-based control system

3.1. Performance analysis

(1) *Disturbance-Rejection Analysis.* Figure 4 is an equivalent block diagram of that in Fig. 2 using transfer functions in the frequency domain. The transfer functions in the figure are

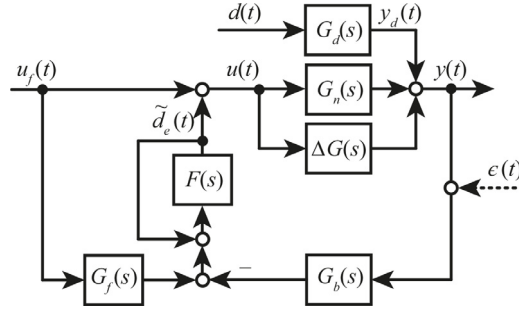


Fig. 4. Equivalent configuration of Fig. 2.

$$\begin{cases} G_n(s) = C(sI - A_n)^{-1}B_n \\ \Delta G(s) = G(s) - G_n(s) \\ G(s) = C(sI - A)^{-1}B \\ G_b(s) = B_n^+ L [1 + C(sI - A_n)^{-1}L]^{-1} \\ G_f(s) = B_n^+ LC [sI - (A_n - LC)]^{-1}B_n \end{cases} \quad (14)$$

where $A = A_n + \Delta A$ and $B = B_n + \Delta B$. The transfer function from $d(t)$ to $y(t)$ is

$$G_{yd}(s) = S(s)G_d(s) \quad (15)$$

where

$$S(s) = \frac{1 - F(s)}{1 - F(s) + F(s)G_b(s)G(s)} \quad (16)$$

is the sensitivity function of the system to disturbances and $G_d(s)$ is the transfer function from $d(t)$ to $y_d(t)$. If τ is chosen to ensure that

$$|F(j\omega)| \approx 1, \quad \forall \omega \in [0, \omega_r] \quad (17)$$

where ω_r is the upper bound of a selected frequency range, then the sensitivity function approximates zero. This means

$$G_{yd}(s) \approx 0 \quad (18)$$

as long as the frequencies of the disturbance are lower than ω_r .

In other words, the disturbance cannot pass through the system if its frequencies stay within the bandwidth of the filter. Thus, it has little effect on the output.

The DOBC method [17] uses a low-pass filter $Q(s)$ to achieve satisfactory disturbance-rejection performance (Fig. 5). The cutoff frequency of $Q(s)$ should be large enough to ensure that the frequencies of a disturbance stay within the bandwidth of the filter. The situation is the same for $F_c(s)$ or $F(s)$ in the EID approach. However, the relative degree of $Q(s)$ in the DOBC method should be equal to or larger than that of a plant to guarantee the implementation of a disturbance observer (DOB) in the method. This makes system design complicated. Although $F_c(s)$ in the conventional EID estimator is simple, the performance is limited by measurement noise. The filter $F(s)$ in the presented IEID estimator provides a way to solve the tradeoff between disturbance rejection and noise suppression.

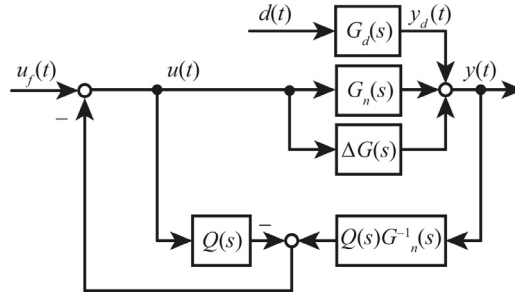


Fig. 5. Conceptual diagram of DOB.

Remark 1. A DOB and an IEID estimator share a similar configuration, that is, an inner loop is designed to estimate a disturbance. However, a DOB uses an inverse model of a plant to calculate an estimate of a disturbance. This might cause the cancellation of unstable poles/zeros. An IEID estimator employs a feedforward ($G_f(s)$) and feedback ($G_b(s)$) mechanism to estimate a disturbance, which avoids the cancellation problem. In Fig. 4, the control input to the plant is

$$u(s) = \frac{1 - F(s) + F(s)G_f(s)}{1 - F(s)}u_f(s) - \frac{F(s)}{1 - F(s)}G_b(s)y(s) \quad (19)$$

where $u(s)$, $u_f(s)$, and $y(s)$ are the Laplace transforms of $u(t)$, $u_f(t)$, and $y(t)$, respectively. In Fig. 5, the input is

$$u(s) = \frac{1}{1 - Q(s)}u_f(s) - \frac{Q(s)}{1 - Q(s)}G_n^{-1}(s)y(s). \quad (20)$$

If $G_f(s) = 1$, $G_b(s) = G_n^{-1}(s)$, and $F(s) = Q(s)$. Clearly, this is exactly a DOB. Thus, an IEID estimator treats a DOB as a special case.

(2) *Uncertainty-Attenuation Analysis.* Let $d(t) = 0$ to analyze the mechanism of uncertainty attenuation. In Fig. 4, the transfer function from $u_f(t)$ to $y(t)$ is

$$G_{yu_f}(s) = G(s)W(s) \quad (21)$$

where

$$W(s) = \frac{1 - F(s) + F(s)G_f(s)}{1 - F(s) + F(s)G_b(s)G(s)}. \quad (22)$$

Since the filter $F(s) \approx 1$ in the bandwidth, one obtains

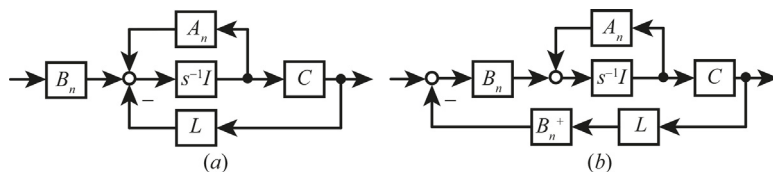
$$W(s) \approx \frac{G_f(s)}{G_b(s)G(s)} = \frac{C[sI - (A_n - LC)]^{-1}B_n}{[1 + C(sI - A_n)^{-1}L]^{-1}C(sI - A)^{-1}B} \quad (23)$$

as the frequency goes into the range $[0, \omega_r]$. The numerator of $W(s)$ can be interpreted by the block diagram in Fig. 6a, which is equivalent to Fig. 6b. Thus, it is easy to obtain

$$C[sI - (A_n - LC)]^{-1}B_n = \frac{C(sI - A_n)^{-1}B_n}{1 + C(sI - A_n)^{-1}L}. \quad (24)$$

Substituting Eq. (24) into Eq. (23) yields

$$W(s) \approx \frac{C(sI - A_n)^{-1}B_n}{C(sI - A)^{-1}B} = \frac{G_n(s)}{G(s)}. \quad (25)$$

Fig. 6. Block diagram for numerator of $W(s)$.

According to Eqs. (21) and (25), we have

$$G_{yuf}(s) \approx G_n(s) \quad (26)$$

as long as the frequency stays in the bandwidth of the filter. This implies that the IEID estimator attenuates parameter uncertainties effectively and the system behaves like a nominal one. On the other hand, if the system has no uncertainties, that is $A = A_n$ and $B = B_n$, then $W(s) = 1$ according to Eqs. (14) and (22). This means that the inner loop does not change the characteristics of the plant as long as $W(s)$ is internally stable.

(3) *Noise-Suppression Analysis.* Add measurement noise $\epsilon(t)$ to the output and let the disturbance be zero in Fig. 4. The transfer functions from $\epsilon(t)$ to $y(t)$ and $\tilde{d}_e(t)$ are

$$G_{y\epsilon}(s) = \frac{F(s)G_b(s)G(s)}{1 - F(s) + F(s)G_b(s)G(s)}, \quad G_{\tilde{d}_e\epsilon}(s) = \frac{F(s)G_b(s)}{1 - F(s) + F(s)G_b(s)G(s)} \quad (27)$$

respectively. If the frequencies of measurement noise stay within the bandwidth of the filter (i.e., Eq. (17) holds), then

$$G_{y\epsilon}(s) \approx 1, \quad G_{\tilde{d}_e\epsilon}(s) \approx \frac{1}{G(s)}. \quad (28)$$

The noise passes through the plant and affects the output. Moreover, the noise is amplified and added to the estimate $\tilde{d}_e(t)$ as $G(s)$ is usually a strictly proper plant. This might cause the malfunction of an actuator and mechanical vibration. Rewrite (27) as

$$G_{y\epsilon}(s) = \frac{G_b(s)G(s)}{\frac{1}{F(s)} - 1 + G_b(s)G(s)}, \quad G_{\tilde{d}_e\epsilon}(s) = \frac{G_b(s)}{\frac{1}{F(s)} - 1 + G_b(s)G(s)}. \quad (29)$$

If we guarantee $|F(j\omega)|$ sufficiently small at high frequencies, then

$$G_{y\epsilon}(s) \approx 0, \quad G_{\tilde{d}_e\epsilon}(s) \approx 0. \quad (30)$$

The disturbance is blocked and the estimate is less sensitive to measurement noise.

Remark 2. In practice, disturbances are usually low or middle frequency and measurement noise is often high frequency. This gives us a way to approach disturbance rejection and noise suppression simultaneously. The filter is designed such that $|F(j\omega)| \approx 0$ at high frequencies and $|F(j\omega)| \approx 1$ over the frequency associated with disturbances. Compared with the conventional EID estimator, a second-order low-pass filter is used to increase the reduction of the magnitude of the filter at high frequencies. This improves noise-suppression performance while keeping disturbance-rejection performance unchanged. On the other hand, if we choose a specified high-frequency band for the filter, it is possible to handle a high-frequency disturbance. Disturbance rejection is directly related to sensitivity suppression and robust stability is related to the suppression of the complementary sensitivity. Since there is a tradeoff between the sensitivity and the complementary sensitivity, we need to pay special attention to this issue

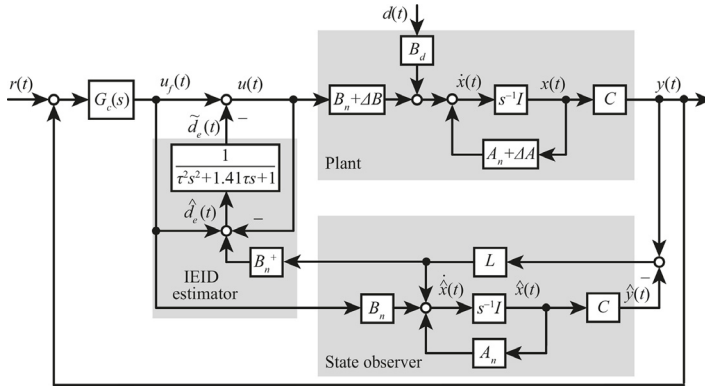


Fig. 7. Configuration of IEID-based control system.

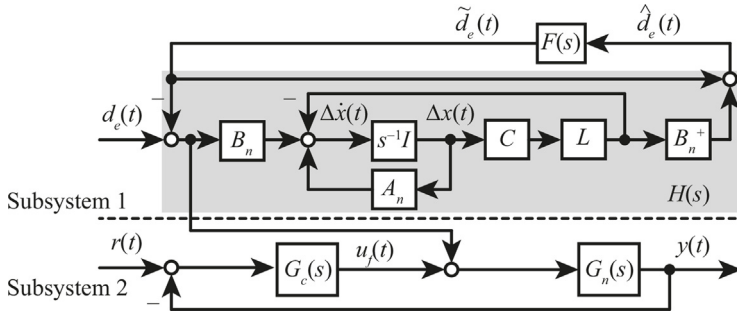


Fig. 8. Equivalent configuration of Fig. 7.

on control system design. For this reason, we usually suppress disturbances at a low or middle frequency and consider robust stability at a high frequency.

3.2. Stability analysis

Feedback control is needed to stabilize the system and achieve satisfactory reference-tracking performance. Since the disturbance and parameter uncertainties are handled by the inner loop, a feedback controller $G_c(s)$ is designed in an outer loop considering the nominal plant $G_n(s)$. The configuration of an IEID-based control system is shown in Fig. 7. The signal $r(t)$ in the figure is a reference input. Simplifying the configuration and redrawing Figs. 7 yield 8. The structure of Subsystem 1 illustrated in Fig. 8 is explained as follows.

The error dynamics of the state observer is

$$\Delta \dot{x}(t) = (A_n - LC)\Delta x(t) + B_n[d_e(t) - \tilde{d}_e(t)] \quad (31)$$

according to Eqs. (3), (8), (9), and (13). Substituting Eq. (13) into Eq. (10) yields

$$\hat{d}_e(t) = B_n^+ LC \Delta x(t) + \tilde{d}_e(t). \quad (32)$$

Moreover, $\tilde{d}_e(t)$ is a filtered value of $\hat{d}_e(t)$. Thus, the structure of Subsystem 1 in Fig. 8 can be obtained from the above equations.

Note that the stability of Subsystem 1 above the dashed line is impervious to Subsystem 2 under the line in Fig. 8. Thus, the stability of the closed-loop system is converted into that of the two subsystems. Define

$$\begin{aligned} H(s) &= 1 - B_n^+ LC[sI - (A_n - LC)]^{-1} B_n \\ &= B_n^+ \{I - LC[sI - (A_n - LC)]^{-1}\} B_n = B_n^+ (sI - A_n)[sI - (A_n - LC)]^{-1} B_n. \end{aligned} \quad (33)$$

According to the small-gain theorem, Subsystem 1 is stable if $H(s)$ and $F(s)$ are both stable, and

$$\|HF\|_\infty < 1 \quad (34)$$

holds. Moreover, $H(s)$ is stable if L is designed such that $A_n - LC$ is Hurwitz. On the other hand, since the nominal plant $G_n(s)$ is known, it is easy to design the controller $G_c(s)$ that stabilizes Subsystem 2. The following theorem is derived to guarantee the stability of the system.

Theorem 1. *The IEID-based closed-loop system is stable if*

- (1) $G_c(s)$ is designed to ensure that Subsystem 2 below the dashed line is stable;
- (2) $A_n - LC$ is Hurwitz;
- (3) $\|HF\|_\infty < 1$.

There are several effective methods of designing the controller $G_c(s)$ for the nominal plant $G_n(s)$ in the literature, for example, the PID control, the state-feedback control, and the pole-zero placement method. Assumption 1 ensures that there exists an L that makes $A_n - LC$ Hurwitz and $[sI - (A_n - LC)]^{-1} B_n \approx 0$ according to the perfect regulation [35]. This indicates that $\|H\|_\infty$ can be made sufficiently small. On the other hand, as uncertainties are treated as a state-dependent disturbance, one may doubt the validity of stability criteria in Theorem 1. Theoretically speaking, a control system has a certain degree of robustness. A small change in a system does not make the system unstable. However, this change may degrade other control performance, for example, tracking. This study considers such a kind of uncertainty and devises an IEID estimator to minimize performance degradation caused by the uncertainty. It has been shown that the uncertainty can be compensated largely as long as its frequency stays within the selected range. In summary, a design procedure is summarized as follows.

- (1) Design the feedback controller $G_c(s)$ for the nominal plant to meet specifications of reference tracking and stability using an existing method.
- (2) Choose the time constant τ of the filter $F(s)$ to make Eq. (17) hold. In general, we set $\tau < 1/(5\omega_r)$.
- (3) Use the perfect regulation method to select an L to make the last two conditions in Theorem 1 hold.

Remark 3. It is intuitive to use a higher-order low-pass filter to achieve noise-suppression performance as it has a larger slope of Bode magnitude curve at high frequencies. It can be seen that the stability criteria are derived based on the small-gain theorem, which focuses on the gain of the filter. Thus, a higher-order low-pass filter has little influence on system stability as long as its amplitude condition $|F(j\omega)| \approx 1$ holds in the bandwidth. On the other hand, a higher-order filter can provide more degrees of regulation freedom. However, this

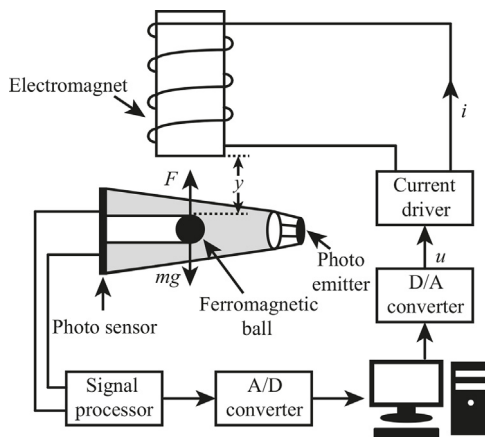


Fig. 9. Schematic diagram of MLS.

might increase the complexity and difficulty of real-time implementation. This is a tradeoff between the effectiveness of disturbance estimation and computational complexity.

Although the above analysis is provided in the frequency domain, it can be extended to the time domain for the following reasons. First, it is easy to obtain the state-space model of the filter $F(s)$. Combining it with Eqs. (3) and (8) yields the state-space model of the closed-loop system in Fig. 2. Then, techniques in the time domain are able to be used to analyze the performance and stability of the system. For example, the Lyapunov method can be used to derive stability criteria and the pole-placement method can be used to specify control performance.

4. Simulation and experimental verification

The position control of an MLS was used to show the design procedure and to verify the validity of the method. Figure 9 shows the schematic diagram of the MLS. A ferromagnetic ball is levitated by magnetic force at a desired position, which is measured by a photo sensor. The parameters in the diagram are given as follows.

m Mass of the ball.

F Magnetic force generated by the electromagnet.

y Position of the ball.

u Control voltage.

i Current in the electromagnet.

g Acceleration of gravity ($g = -9.8 \text{ m/s}^2$).

The dynamics of the ball is

$$m \frac{d^2 y}{dt^2} = F(i, y) + mg. \quad (35)$$

The magnetic force F is determined by the current i and the position y . It is given by

$$F(i, y) = k_f \left(\frac{i}{y} \right)^2 \quad (36)$$

where k_f is an electromagnetic coefficient. Linearizing the dynamics around an equilibrium point yields

$$\ddot{y}(t) = \frac{2g}{y_0} y(t) - \frac{2g}{i_0 k_a} u(t) \quad (37)$$

where y_0 and i_0 are the position and the current of the MLS at the equilibrium point, respectively; and k_a is the gain of the current driver. Let $x(t) = [y(t), \dot{y}(t)]^T$ be the state vector. The state-space model of Eq. (37) is

$$\begin{cases} \dot{x}(t) = Ax(t) + Bu(t) \\ y(t) = Cx(t) \end{cases} \quad (38)$$

where

$$A = \begin{bmatrix} 0 & 1 \\ \frac{2g}{y_0} & 0 \end{bmatrix}, \quad B = \begin{bmatrix} 0 \\ -\frac{2g}{i_0 k_a} \end{bmatrix}, \quad C = [1, \quad 0].$$

The objective position was set to be -10 mm. The parameters of the experimental setup were identified to be

$$y_0 = -10 \text{ mm}, \quad i_0 = 0.61 \text{ A}, \quad k_a = 5.89. \quad (39)$$

A PID controller

$$G_c(s) = K_P + \frac{K_I}{s} + K_D s \quad (40)$$

was used in the outer loop to stabilize the MLS. The gains of the controller were set to be

$$K_P = 750, \quad K_I = 450, \quad K_D = 135 \quad (41)$$

by trial and error. The disturbance

$$d(t) = 0.5 \sin(\pi t) + 0.3 \cos(0.3\pi t) \quad (42)$$

was added to the MLS during $5 - 15$ s. We set

$$\tau = 0.02 \text{ s} \quad (43)$$

to ensure that Eq. (17) holds in the frequency band of the disturbance.

The perfect regulation method was employed to choose the gain of the state observer. For the dual system (A^T, C^T, B^T) , find an L to minimize the performance index

$$J_L = \int_0^\infty [\rho x_L^T(t) Q_L x_L(t) + R_L u_L^2(t)] dt \quad (44)$$

$$Q_L = \text{diag}\{1 \quad 10^5\}, \quad R_L = 1, \quad u_L(t) = L^T x_L(t)$$

where $x_L(t)$ was the state of the dual system and ρ is a tuning parameter. As ρ increased, $|H(j\omega)|$ decreased. Selecting $\rho = 10^4$ yielded

$$L = [278 \quad 33643]^T. \quad (45)$$

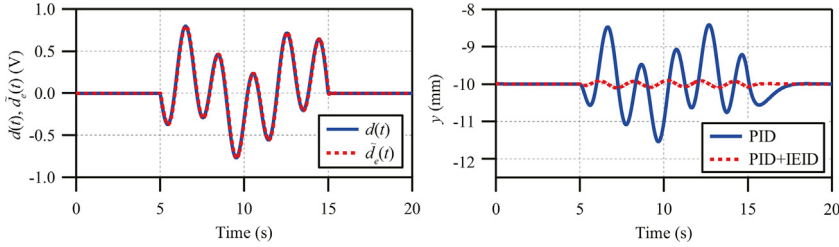


Fig. 10. Disturbance estimation and rejection for MLS without uncertainty.

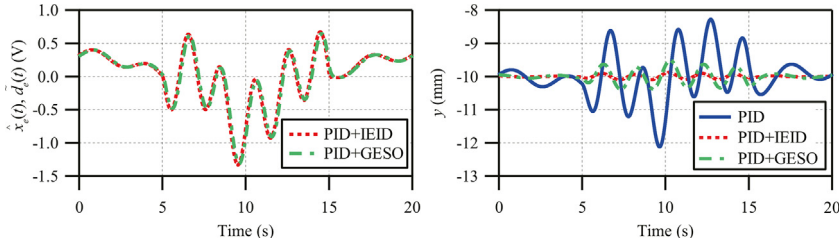


Fig. 11. Disturbance rejection and uncertainty attenuation for IEID and GESO methods.

It is easy to verify that the last two conditions in [Theorem 1](#) are satisfied. The MATLAB *lqr* function was used to solve the observer gain. The solver *ode3* was chosen for simulations and experiments. Since the performance of disturbance rejection was mainly focused, we assumed that the MLS had already entered in steady state at the initial time $t = 0$ s.

4.1. Simulation results

First, the parameter uncertainties were neglected to investigate the performance of disturbance estimation and rejection. The disturbance [Eq. \(42\)](#) was added to the system on the control input channel. [Figure 10](#) shows that the disturbance is estimated precisely and has little effect on the output for the control with the IEID estimator. The outer loop PID control stabilized the MLS and achieved satisfactory positioning performance when the disturbance was not imposed on. However, it can be seen that its disturbance-rejection capacity is limited. Compared to the outer loop, the inner loop is an effective way to handle exogenous disturbances.

The linearized model was obtained using Taylor's series around an equilibrium point. Thus, the model parameters changed as the operating point deviated from the equilibrium one. Moreover, changes in resistance and inductance caused by electromagnet heating also caused parameter variations. The uncertainties in A and B to be tested in simulations were

$$\Delta A = \begin{bmatrix} 0 & 0 \\ 50 \sin(0.6\pi t) & 0 \end{bmatrix}, \quad \Delta B = \begin{bmatrix} 0 \\ 0.55 \sin(0.1\pi t) \end{bmatrix}. \quad (46)$$

The simulation results show that the disturbance and the uncertainties are well estimated as the effect of them on the output is reduced largely ([Fig. 11](#)). The simulations were also carried out for the generalized extended state observer (GESO)-based PID control [\[36\]](#). First, an extended state was used to denote the disturbance and the uncertainties. Then, a GESO was employed to estimate it in an inner loop. For a fair comparison, the bandwidth of the

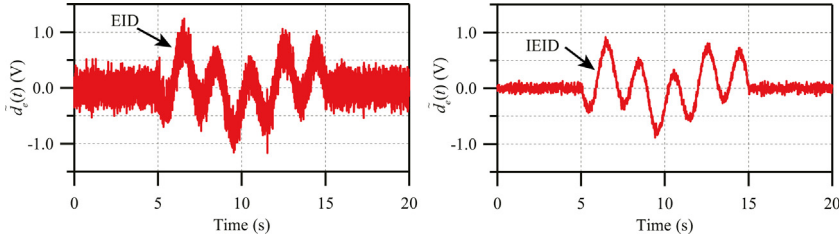
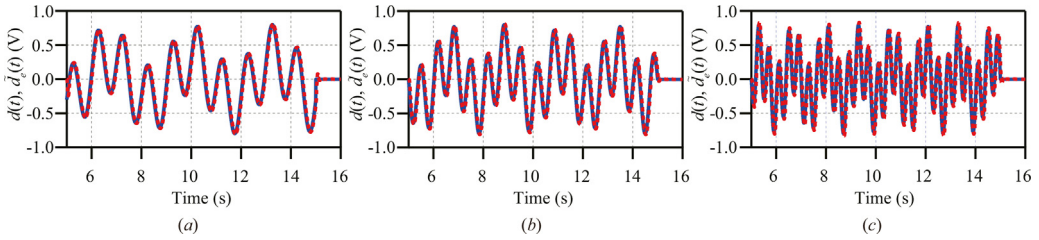


Fig. 12. Noise suppression for IEID and EID estimators.

Fig. 13. Estimation performance of various frequency disturbance inputs (solid line: disturbance, dotted line: estimate): (a) $\omega_r = 2\pi$, (b) $\omega_r = 3\pi$, and (c) $\omega_r = 5\pi$.

GESO was set to be not less than that of the IEID estimator. The gain of the GESO was chosen as

$$L_g = [290 \ 36400 \ 316230]^T. \quad (47)$$

The estimate of the extended state is described by $\hat{x}_e(t)$ in Fig. 11. The peak-to-peak value of the output error is 3.85 mm for the PID control without the inner loop and 0.88 mm for the GESO-based PID control. It is 0.22 mm for the IEID-based PID control, which is 5.8% and 25% of those for the PID control and the GESO-based PID control, respectively. This reveals that the presented method is more effective in handling disturbances and uncertainties than the other two methods.

We added measurement noise to the output to show the superiority of the IEID estimator over the conventional EID estimator in noise suppression. The noise was produced by the Simulink block “Band-Limited White Noise”. The noise power and the sampling time were set to be 10^{-10} and 0.001 s, respectively. The magnitude of the noise was bounded in $[-0.001 \ 0.001]$. For the conventional EID estimator, the filter $F_c(s)$ was chosen and the time constant was also set to be 0.02 s. The decline rate of the magnitude curve at high frequencies was two times larger for $F(s)$ (in the IEID estimator) than for $F_c(s)$ (in the EID estimator). Thus, the estimate was less sensitive to the noise for the IEID estimator than for the EID estimator (Fig. 12). This implies that the IEID estimator is more practical.

The frequencies of the disturbance Eq. (42) were multiplied by 2, 3, and 5. Figure 13 shows estimation performance. The highest angular frequency is 2π rad/s in Fig. 13(a), 3π rad/s in Fig. 13(b), and 5π rad/s in Fig. 13(c). As long as frequencies stayed within the bandwidth of the filter, the estimate tracked the disturbance effectively. It can be seen that the more the cutoff frequency of the filter is greater than the maximum frequency of the disturbance, the better the estimation performance will be.

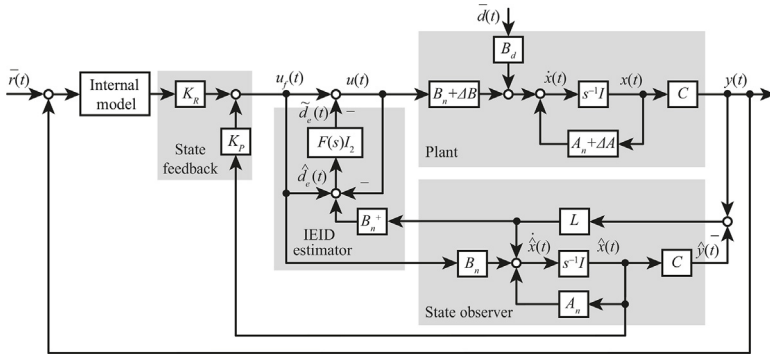


Fig. 14. Configuration of the IEID-based control system for dual-stage feed drive.

We extended the presented method to a multi-input-multi-output (MIMO) system. A dual-stage feed drive control system was studied. The dynamic model was derived by She et al. [23]. Here, the system matrices are given directly

$$A_n = \begin{bmatrix} 0 & 1 & 0 & 0 \\ -120 & -12 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 20 & 2 & 0 & 0 \end{bmatrix}, \quad B_n = \begin{bmatrix} 0 & 0 \\ 60 & -10 \\ 0 & 0 \\ -10 & 10 \end{bmatrix}, \quad C_n = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}. \quad (48)$$

Figure 14 shows the configuration of the control system. The internal model and state feedback control (SFC) were used instead of the PID control. A step reference $r(t) = 1$ was added to the system. A low-pass filter

$$F_r(s) = \frac{1}{0.2s + 1} \quad (49)$$

was employed to generate two reference inputs for the two stages

$$\bar{r}(t) = \begin{bmatrix} F_r(s) \\ 1 - F_r(s) \end{bmatrix} r(t). \quad (50)$$

Since the reference was a step signal, the internal model was selected to be an integrator. The gains of the state feedback were chosen to be

$$K_R = \begin{bmatrix} 0.98978 & 0.14260 \\ -0.14260 & 0.98978 \end{bmatrix}, \quad K_P = \begin{bmatrix} -29.394 & -0.81711 & -0.21541 & -0.0908 \\ 4.5192 & 0.02207 & -1.4680 & -0.5939 \end{bmatrix} \quad (51)$$

using the linear-quadratic-regulator method. The gain of the observer was designed as

$$L = \begin{bmatrix} 1402.179 & 983053.6 & 0.998572 & 2117.242 \\ 0.998572 & 695.1309 & 1414.214 & 1000002 \end{bmatrix}^T \quad (52)$$

by the perfect-regulation method. The disturbance $\bar{d}(t) = [d_1(t) \ d_2(t)]^T$ was imposed during 5 – 15 s

$$d_1(t) = 0.2(\sin 5t + 0.5 \tanh 14t + \cos 8t), \quad d_2(t) = 0.4 \tanh(t - 5) - 0.2 \tanh t + 0.1 \sin(2\pi t). \quad (53)$$

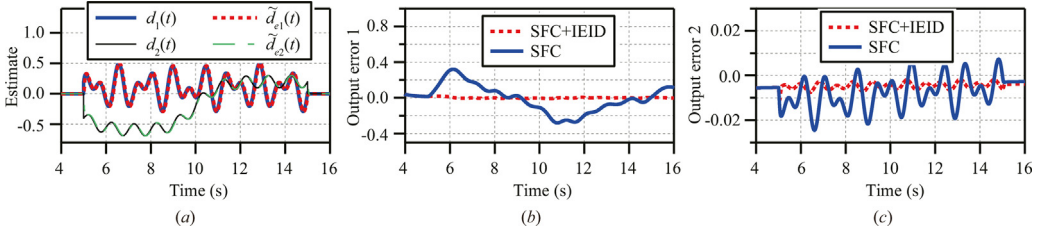


Fig. 15. Simulation results of disturbance estimation and rejection for dual-stage feed drive control system.

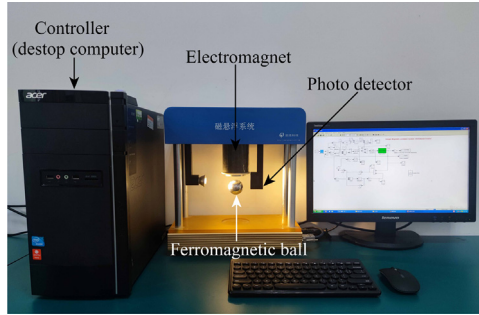


Fig. 16. Experimental setup.

The same filter $F(s)$ ($\tau = 0.02$ s) was used.

Figure 15 shows simulation results of disturbance estimation and rejection. Since the cut-off angular frequency of the filter is much larger than frequencies of the disturbances, the estimates $\hat{d}_{e1}(t)$ and $\hat{d}_{e2}(t)$ approaches to the disturbances $d_1(t)$ and $d_2(t)$, respectively. Alternatively, different filters can be chosen in the IEID estimator for a MIMO system to handle disturbances on different channels. The same filters were used in the dual-stage feed drive system for simplicity. The peak-to-peak values of the system output errors were 0.0318 and 0.6001 during 5–15 s for the conventional state feedback control. They reduced to 0.0092 and 0.0286 for the IEID-based control. It indicates that the effect of the disturbances on the system outputs was greatly reduced by incorporating the estimates into the control law.

4.2. Experimental results

The experimental setup of the MLS is shown in Fig. 16. The control program was implemented using a desktop computer with the Real-Time Toolbox of MATLAB. The sampling time was set to be 3 ms. The parameters of the setup are shown in (39). We added a 20% uncertainty in A and a 10% uncertainty in B , that is

$$\Delta A = \begin{bmatrix} 0 & 1 \\ 392 & 0 \end{bmatrix}, \quad \Delta B = \begin{bmatrix} 0 \\ 0.55 \end{bmatrix}. \quad (54)$$

The same PID controller, the observer gain L , and the filter $F(s)$ were used in experiments.

Figure 17 shows that the estimate $\hat{d}_e(t)$ recovers the trajectory of the disturbance plus an extra constant component, which represents the estimate of uncertainties on the control input channel. The system was robust to the disturbance and uncertainties as the output was nearly unaffected by them. Moreover, the outer-loop control input $u_f(t)$ was smaller for the control

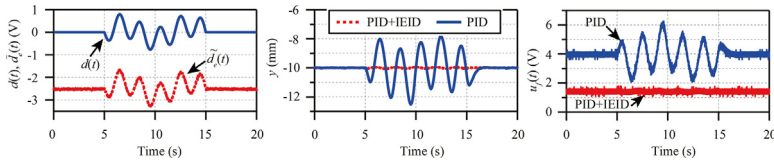


Fig. 17. Experimental results for MLS with disturbance and uncertainty.

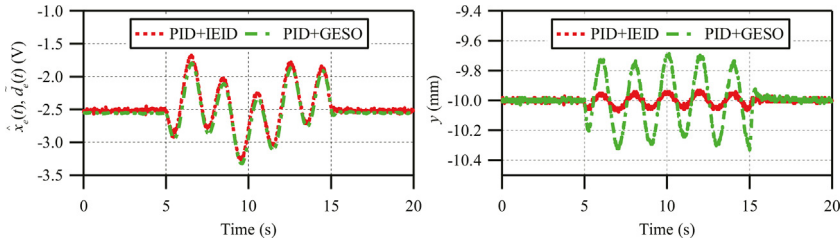


Fig. 18. Experimental results for the IEID and GESO methods.

with the IEID estimator than for the PID control. This indicates that the performance of disturbance rejection and uncertainty attenuation is mainly achieved by the inner loop (IEID estimator). Since the derivative control was used in the outer loop, the control law $u_f(t)$ was sensitive to measurement noise. In contrast, the estimate $\tilde{d}_e(t)$ was less sensitive to the noise as a low-pass filter was employed in the inner loop.

The gain L_g in Eq. (47) was also used in experiments for the GESO-based PID control. Figure 18 shows that the output error is almost zero outside the period 5–15 s. This implies that the constant uncertainties are completely counteracted by both methods. The estimate $\tilde{d}_e(t)$ is a better one as the output error is smaller for the IEID-based PID control than for the GESO-based PID control during the period. The IEID approach is more effective in rejecting disturbances.

5. Conclusion

This paper has developed a robust control structure that employed an inner loop to handle exogenous disturbances and parameter uncertainties based on the EID approach. The IEID estimator that used a second-order low-pass filter was presented to filter out high components of measurement noise and thus to improve noise-suppression performance. The mechanism of disturbance rejection and uncertainty attenuation for the IEID-based control system was analyzed in the frequency domain. Satisfactory performance was guaranteed as long as frequencies of disturbances and uncertainties stayed in the bandwidth of the filter. It was shown that the inner loop was a more effective way of rejecting disturbances and uncertainties than the outer loop. Simulation and experimental results of the position control of the MLS demonstrated the robustness and effectiveness of the presented method. It can be seen that the filter plays a key role in rejecting disturbances and suppressing measurement noise. It is important to carry out quantitative research on the design of the filter. An alternative way to improve noise-suppression performance is to use an observer with a time-varying gain, for example, an extended state observer in [37] and a high-gain observer in [38]. Future work will be focused on these topics.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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